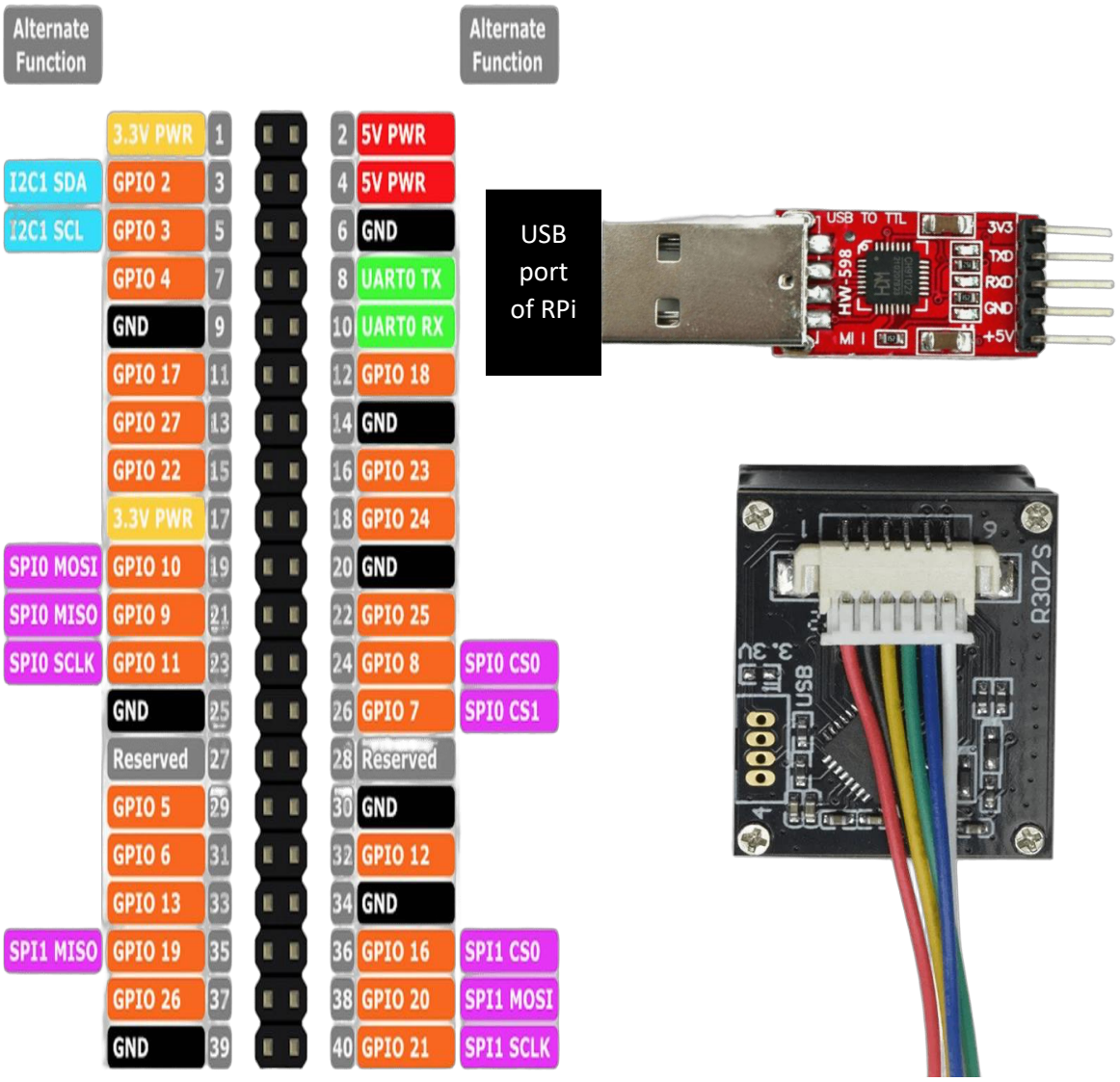
PRACTICAL 8 (Module 1)

TITLE : Interface Fingerprint Sensor with Raspberry Pi to read and write to RFID tags/cards

- **AIM :** To interface Fingerprint sensor with Raspberry Pi and to enroll, find and delete the fingerprint.
- **HARDWARE COMPONENTS REQUIRED :** Raspberry Pi 400 (Pi version 4), power adaptor, HDMI to micro HDMI cable, mouse, monitor, Fingerprint sensor module (R307), USB to TTL UART Converter Module (CP2102), jumper wires
- **CIRCUIT SCHEMATIC :**



NOTE : Draw and show the connections of the USB to TTL converter with Fingerprint sensor module as per the reference table given below -

(Draw the exact same diagrams for the raspberry pi and the modules as the image attached aside with all the pin numbers and names)

Fingerprint sensor (R307) module pin number	**Fingerprint sensor (R307) module pin number**	**USB to TTL converter pins**
1	Vcc	3V / 5V
2	Gnd	Gnd
3	Tx	Rx
4	Rx	Tx

- **COMMANDS :**
  - sudo apt-get update (Updates the local list of available software packages)
  - sudo apt-get upgrade (Updates software packages installed on linux system)
  - sudo raspi-config (Enable SPI)
  - lsmod | grep spi (Checks, loads and runs the SPI drivers)
  - sudo apt -get install python3-dev python3-pip (Installs python 3 packages required)
  - sudo pip3 install spidev (Installs python 3 modules required to communicate with the SPI devices)
  - sudo pip3 install mfrc522 (Installs MFRC522 python library)

- **PROGRAM :**

```
import time
import board
import adafruit_fingerprint
import serial
uart = serial.Serial("/dev/ttyUSB0", baudrate=57600)
finger = adafruit_fingerprint.Adafruit_Fingerprint(uart)

def get_fingerprint():
    print("Waiting for image...")
```

```
    while finger.get_image() != adafruit_fingerprint.OK:
        pass
    print("Templating...")
    if finger.image_2_tz(1) != adafruit_fingerprint.OK:
        return False
    print("Searching...")
    if finger.finger_search() != adafruit_fingerprint.OK:
        return False
    return True


def enroll_finger(location):
    for fingerimg in range(1, 3):
        if fingerimg == 1:
            print("Place finger on sensor...", end="")
        else:
            print("Place same finger again...", end="")
        while True:
            i = finger.get_image()
            if i == adafruit_fingerprint.OK:
                print("Image taken")
                break
            elif i == adafruit_fingerprint.NOFINGER:
                print(".", end="")
            else:
                print("Other error")
                return False
        print("Templating...", end="")
        i = finger.image_2_tz(fingerimg)
        if i == adafruit_fingerprint.OK:
            print("Templated")
        else:
            print("Other error")
            return False
        if fingerimg == 1:
            print("Remove finger")
            time.sleep(1)
            while i != adafruit_fingerprint.NOFINGER:
                i = finger.get_image()
    print("Creating model...", end="")
    i = finger.create_model()
    if i == adafruit_fingerprint.OK:
        print("Created")
    else:
            print("Error")
            return False
    print(f"Storing model #{location}...", end="")
    i = finger.store_model(location)
    if i == adafruit_fingerprint.OK:
        print("Stored")
    else:
```

```
        print("Error!")
        return False
    return True


def get_num():
    i = 0
    while (i > 127) or (i < 1):
        try:
            i = int(input("Enter ID # from 1-127: "))
        except ValueError:
            pass
    return i


while True:
    if finger.read_templates() != adafruit_fingerprint.OK:
        raise RuntimeError("Failed to read templates")
    print("Fingerprint templates:", finger.templates)
    print("e - To enroll")
    print("f - To find")
    print("d - To delete")
    c = input("> ")

    if c == "e":
        enroll_finger(get_num())
        print("Stored successfully!")
    if c == "f":
        if get_fingerprint():
            print("Detected #", finger.finger_id)
        else:
            print("Finger not found")
    if c == "d":
        if finger.delete_model(get_num()) == adafruit_fingerprint.OK:
            print("Deleted!")
        else:
            print("Failed to delete")
```

- **RESULS & CONCLUSION** : **Note –** Write the output displayed on the Serial monitor (If no output got - Mention the errors displayed)